Chapter 2

Matrix Algebra



MSc. Nguyen Cam Nhiem
Email: nhiemnc@fe.edu.vn

Contents

2.1 Matrix Addition, Scalar Multiplication, and Transposition

2.2 Matrix Multiplication (2.3 7e)

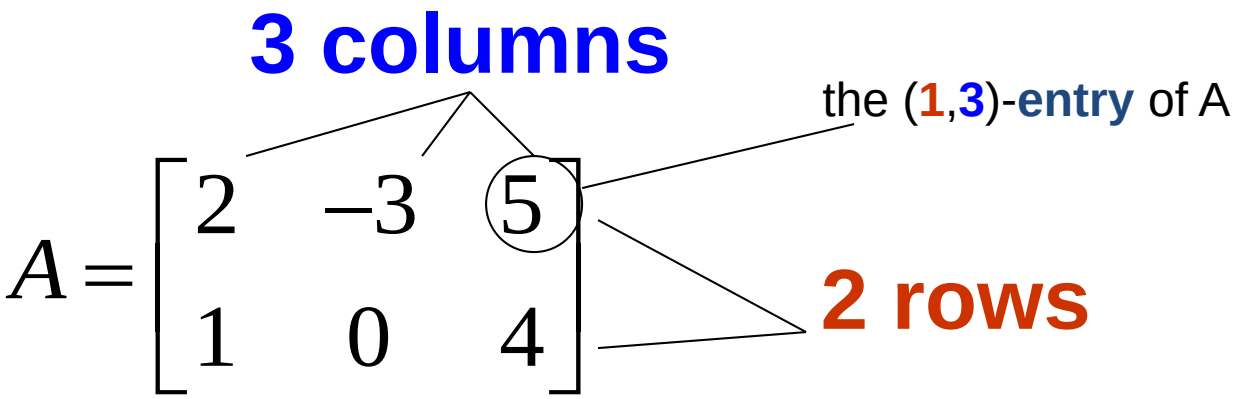
2.3 Matrix Inverses (2.4 7e)

2.5 Matrix Transformations (2.2 & 2.6 7e)

2.1. Matrix Addition, scalar multiplication and transposition

Definitions

- An **m**x**n** matrix (or a matrix of size **m**x**n**) is a rectangular array of numbers with **m rows** and **n columns**



A is a **2**x**3** matrix (or a matrix of size **2**x**3**)

The mxn matrix

- The (**i**,**j**)-entry of A (denoted by a_{ij}) lies in **row i** and **column j**
- A is denoted simply as A=[a_{ij}]

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

2 refers to the **row**

n refers to the **column**

Definitions

- An **mxm** matrix is called a **square matrix (ma trận vuông)** of size m.
- If $A=[a_{ij}]$ is any mxn matrix, then $\mathbf{a_{11}, a_{22}, a_{33}}$,…, are called the **main diagonal (đường chéo chính)** of A.
- The **zero matrix (ma trận không)** of size mxn (denoted by $(0_{mxn})$is the matrix that its all entries are 0
- If $A=[a_{ij}]$ is an mxn matrix then **-A** refers to the **negative matrix (ma trận đối)** of A and defined by

$$\mathbf{-A=[-a_{ij}]}$$

Identity matrices (ma trận đơn vị)

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \ldots$$

An **identity matrix I** is a square matrix with 1's on the main diagonal and zeros elsewhere

Triangular matrices

- **Upper triangular** matrix: all entries **below** and **to the left** the **main diagonal** are zeros
- **Lower triangular** matrix: its transposition is upper triangle matrix, that means every entry above and to the right the main diagonal is zero
- Matrix A is called **triangular** if it is upper or lower triangular
- For example, every **row-echelon matrix** is upper triangular

$$U = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -7 & 8 & 5 & 0 \end{bmatrix}$$

Two matrices are called **equal** if

- They have the same size
- Corresponding entries are equal

If $A=[a_{ij}]$, $B=[b_{ij}]$ then **$A=B$** means **$a_{ij}=b_{ij}$** for all i and j

Example 1: Given $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B=\begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \end{bmatrix}$, $C=\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$, discuss the possibility that $A=B, B=C, A=C$.

Matrix Addition of same size matrices

- If $A=[a_{ij}]$, $B=[b_{ij}]$ then the sum matrix A+B is defined by $\mathbf{A+B=[a_{ij}+b_{ij}]}$
- The **difference A-B** is a matrix defined by

  $\mathbf{A-B=A+(-B)=[a_{ij}-b_{ij}]}$ for all mxn matrices A and B

  Note that A-A=0, A+0=A (0 is zero matrix) for all mxn matrix A

Example 2: If $A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 6 \end{bmatrix}$, compute A+B.

Example 3: Find a, b and c if $\begin{bmatrix} a & b & c \end{bmatrix} + \begin{bmatrix} c & a & b \end{bmatrix} = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$

Properties

If A ,B and C are any matrices of the same size, then

- A+B=B+A (commutative law: giao hoán)
- A+(B+C)=(A+B)+C (associative law: kết hợp)

Example 5: Solve $\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} + X = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$, where X is a matrix.

Scalar Multiplication
(phép nhân vô hướng)

- Suppose $\mathbf{A=[a_{ij}]}$ is an mxn matrix and **k** is a **real munber**, the **scalar multiple kA** is a matrix defined by $\mathbf{kA=[ka_{ij}]}$
- $kA=\mathbf{0}\rightarrow$ (either $k=0$ or $A=\mathbf{0}$)
- ($k=0$ or $\mathbf{A=0}$)$\rightarrow\mathbf{kA=0}$

Scalar: a quantity that has magnitude, but not direction; -- distinguished from a vector, which has both magnitude and direction (Webster Dictionary) → **One value**

Example 6:

If $A=\begin{bmatrix} 3 & -1 & 4 \\ 2 & 0 & 6 \end{bmatrix}, B=\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$. Compute $5A, \frac{1}{2}B, 3A-2B$.

Theorem 1

Let A, B and C denoted arbitrary $m \times n$ matrices where m and n are fixed.

Let k and p denoted arbitrary real numbers. Then

1. $A + B = B + A$
2. $A + (B + C) = (A + B) + C$
3. There is an $m \times n$ matrix 0, such that $0 + A = A$ for each A.
4. For each A there is an $m \times n$ matrix, $-A$, such that $A + (-A) = 0$.
5. $k(A + B) = kA + kB$.
6. $(k + p)A = kA + pA$.
7. $(kp)A = k(pA)$.
8. 1A=A.

Transpose

- If $A=[a_{ij}]$ is any **m**x**n** matrix,the **transpose** of A, written $\mathbf{A^T}$, is an **n**x**m** matrix defined by $A^T=[a_{ji}]$

- The row i of A is the column i of A^T
- The column j of A is the row j of A^T

Example 9: Write down the transpose of each of the following matrices.

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 2 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$$

Theorem 2

- $(A^T)^T=A$
- $(kA)^T=k(A^T)$
- $(A+B)^T=A^T+B^T$

Definitions

- If $A=[a_{ij}]$ is any mxn matrix, then **$a_{11}, a_{22}, a_{33}, \ldots$, are called the main diagonal (đường chéo chính)** of A
- If **$A=A^T$** then A is called **symmetric (đối xứng).** In this case, A is a **square matrix**

Dot product (tích vô hướng)

Example

$$\begin{pmatrix}1 & 2 & 3\end{pmatrix}\cdot\begin{pmatrix}3\\2\\1\end{pmatrix}=1\times3+2\times2+3\times1=10$$

2.2 Matrix Multiplication

- Suppose $A=[a_{ij}]$ is an **m**x**k** matrix and $B=[b_{ij}]$ is an **k**x**n** matrix, then the **product $AB=[c_{ij}]$** is an **m**x**n** matrix whose the **(i,j)-entry** is the **dot product** of **row i** of A and **column j** of B
- c_{ij}=(**row i** of A).(**column j** of B)
- Note that $A_{mxk}B_{kxn}$ is a **m**x**n** matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \\ -2 & 0 \end{pmatrix} \times \begin{pmatrix} 1 & 0 & -1 & 2 \\ 1 & 2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1.\ 3\cdot 2\ 4 & 1 & 2 \\ -1 & -2 & -1 & 0 \\ -2 & 0 & 2 & -4 \end{pmatrix}$$

Theorem 1

Assume that k is an arbitrary scalar and A, B and C are matrices of sizes such that the indicated operations can be performed.

1. $IA = A,\ BI = B$
2. $A(BC) = (AB)C$
3. $A(B+C) = AB + AC; A(B-C) = AB - AC$
4. $(B+C)A = BA + CA; (B-C)A = BA - CA$
5. $k(AB) = (kA)B = A(kB)$
6. $(AB)^T = B^T A^T$.

Example

$$A=\begin{bmatrix}1 & -1\\ 0 & 0\end{bmatrix};B=\begin{bmatrix}a & b\\ 1 & 0\end{bmatrix}$$

Find B such that:

- B **commutes** with A
- B^2 is **zero matrix**

Matrix and system of linear equations

- Consider the system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m \end{cases}$$

If $A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$, these equations become the single matrix equation

$$AX = B$$

Consider the following system equations:

$$3x_1 - 2x_2 + x_3 = b_1$$
$$2x_1 + x_2 - x_3 = b_2$$

We write $A = \begin{bmatrix} 3 & -2 & 1 \\ 2 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, the bove system of equations becomes the single matrix equation $AX = B$

Example 8: Solve the homogeneous system $AX = 0$ where $A = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -3 & 6 & 1 & 0 \\ -2 & 4 & 4 & -2 \end{bmatrix}$.

Solution

The reduction of the augmented matrix to reduced form is

$$\left\langle \begin{array}{cccc|c} 1 & -2 & 3 & -2 & 0 \\ -3 & 6 & 1 & 0 & 0 \\ -2 & 4 & 4 & -2 & 0 \end{array} \right\rangle \rightarrow \left\langle \begin{array}{cccc|c} 1 & -2 & 0 & -\frac{1}{5} & 0 \\ 0 & 0 & 1 & -\frac{3}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right\rangle$$

So the solutions are

$$x_1 = 2s + \frac{1}{5}t$$
$$x_2 = s$$
$$x_3 = \frac{3}{5}t$$
$$x_4 = t$$

by Gaussian elimination.

two parameters

Hence we can write the general solution X in the matrix form

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2s + \frac{1}{5}t \\ s \\ \frac{3}{5}t \\ t \end{bmatrix} = s\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix} = sX_1 + tX_2$$

linear combination of X_1, X_2

basic solutions

where $X_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \\ 1 \end{bmatrix}$ are particular solutions determined by the Gaussian algorithm.

Example 10: Find basic solutions of the system $AX = 0$ and express every solutions as a linear combination of the basic solutions, where

$$A = \begin{bmatrix} 1 & -3 & 0 & 2 & 2 \\ -2 & 6 & 1 & 2 & -5 \\ 3 & -9 & -1 & 0 & 7 \\ -3 & 9 & 2 & 6 & -8 \end{bmatrix}$$

Solution

The reduction of the augmented matrix to reduced row – echelon form is

$$\left\langle\begin{array}{rrrrr|c} 1 & -3 & 0 & 2 & 2 & 0 \\ -2 & 6 & 1 & 2 & -5 & 0 \\ 3 & -9 & -1 & 0 & 7 & 0 \\ -3 & 9 & 2 & 6 & -8 & 0 \end{array}\right\rangle \rightarrow \left\langle\begin{array}{rrrrr|c} 1 & -3 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 6 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 & 0 \end{array}\right\rangle$$

so the general solution is

$$x_1 = 3r - 2s - 2t$$
$$x_2 = r$$
$$x_3 = -6s + t$$
$$x_4 = s$$
$$x_5 = t$$

, where r, s, t are parameters. In matrix form this is

basic solutions

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3r - 2s - 2t \\ r \\ -6s + t \\ s \\ t \end{bmatrix} = r\begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Theorem 3

Let A be an mxn matrix of **rank r**, **AX=0** is an homogeneous in **n** variables. Then:

- The system has exactly **n-r basic solutions**, one for each parameter.
- Every solution is a **linear combination (tổ hợp tuyến tính)** of these basic solutions.

basic solutions

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3r - 2s - 2t \\ r \\ -6s + t \\ s \\ t \end{bmatrix} = r\begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} -2 \\ 0 \\ -6 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

linear combination

Block Multiplication

If A is a $m \times n$ and $C_1, C_2, \ldots, C_n$ are the columns of A, we write

$$A = \begin{bmatrix} C_1 & C_2 \ldots C_n \end{bmatrix}.$$

Theorem 4

Let $A = \begin{bmatrix} C_1 & C_2 \ldots C_n \end{bmatrix}$ be an $m \times n$ matrix with columns $C_1, C_2, \ldots, C_n$. If $X = \begin{bmatrix} x_1 & x_2 \ldots x_n \end{bmatrix}^T$ is any column, then

$$AX = \begin{bmatrix} C_1 & C_2 \ldots C_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 C_1 + x_2 C_2 + \ldots + x_n C_n .$$

Theorem 5

Suppose that matrices $A = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$ and $A_1 = \begin{bmatrix} B_1 & X_1 \\ 0 & C_1 \end{bmatrix}$ are partitioned as shown where B and B_1 are square matrices of the same size, and C and C_1 are square of the same size. These are compatible partitionings and block multiplication gives

$$AA_1 = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}\begin{bmatrix} B_1 & X_1 \\ 0 & C_1 \end{bmatrix} = \begin{bmatrix} BB_1 & BX_1 + XC_1 \\ 0 & CC_1 \end{bmatrix}$$

As an example, consider the matrices

$$A=\left[\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \hline 2 & -1 & 4 & 2 & 1 \\ 3 & 1 & -1 & 7 & 5 \end{array}\right]=\begin{bmatrix} I_2 & 0_{23} \\ P & Q \end{bmatrix} \text{ and } B=\left[\begin{array}{cc} 4 & -2 \\ 5 & 6 \\ \hline 7 & 3 \\ -1 & 0 \\ 1 & 6 \end{array}\right]=\begin{bmatrix} X \\ Y \end{bmatrix}$$

The product AB can be computed in blocks as follows:

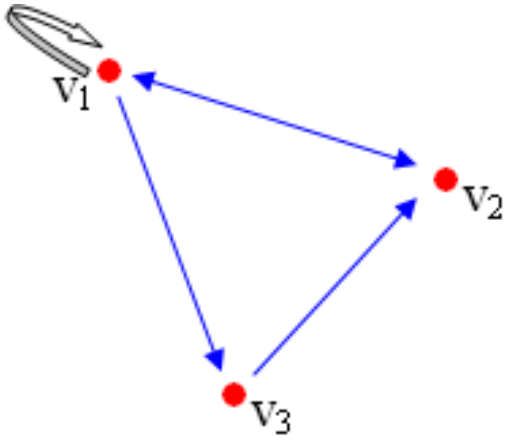
$$AB=\begin{bmatrix} I & 0 \\ P & Q \end{bmatrix}\begin{bmatrix} X \\ Y \end{bmatrix}=\begin{bmatrix} IX+0Y \\ PX+QY \end{bmatrix}=\begin{bmatrix} X \\ PX+QY \end{bmatrix}=\begin{bmatrix} 4 & -2 \\ 5 & 6 \\ 30 & 8 \\ 8 & 27 \end{bmatrix}$$

Exercises 11, 12 p67

Directed Graph

- Points=vertices
- Vertices are connected by **arrows** called **edges**
- If a directed graph has **n vertices** $v_1, v_2, \ldots, v_n$, the **adjacency matrix** $A=[a_{ij}]$ is the **nxn** matrix
- The (i,j)-entry is only **1** or **0**
- If there is an edge from v_j to vertex v_i then $a_{ij}=1$ else $a_{ij}=0$

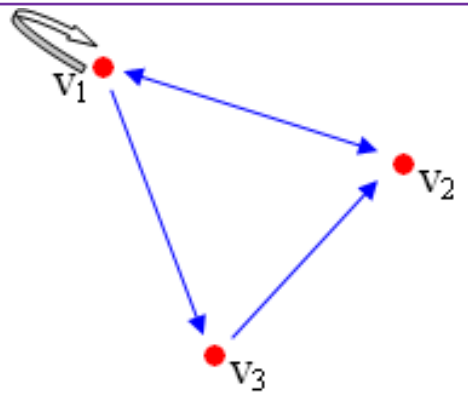
Directed Graph



- a_{11}=1 means $v_1 \rightarrow v_1$
- a_{23}=1 means $v_3 \rightarrow v_2$
- a_{32}=0 means have no edges from vertex **2** to vertex **3**
- A **path of length r** (or an r-path) from vertex **j** to vertex **i** is a sequence of r edges leading from v_j to v_i.
- $v_1 \rightarrow v_2 \rightarrow v_1 \rightarrow v_3 \rightarrow v_2$ is a **4-path** from v_1 to v_2

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Theorem 6



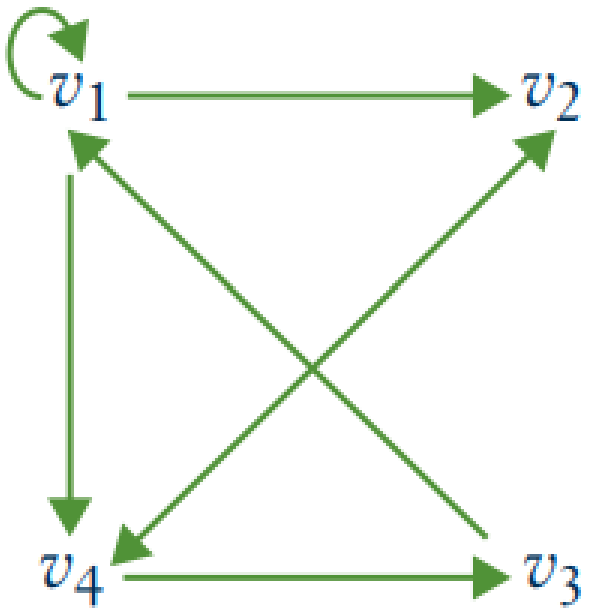
- If A is the adjacency matrix of a directed graph with n vertices, then the (**i**,**j**)-entry of A^r is the number of **r-paths $v_j \rightarrow v_i$**
- For example, in **A^3**, (**2**,**1**)-entry is **3**, so there are **three 3-paths** from **v_1** to **v_2**
- **There are one 3-paths from v_2 to v_3 as (3,2)-entry of A^3 is 1**

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 4 & 2 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

Exercises 26 - p68

For the directed graph at the right, find the adjacency matrix A, compute A^3, and determine the number of paths of length 3 from v_1 to v_4 and from v_2 to v_3.



2.3 Matrix Inverse

Definition

- If A is a **square matrix**, a matrix B is called an **inverse** (nghịch đảo) of A if and only if **AB=I** and **BA=I**
- A matrix A that has an inverse is called an **invertible (khả nghịch)** matrix.

Example

- Show that $A^3=I$ and find A^{-1} if

$$A=\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$$

- Solve the system AX=B with $B=[1\ \ 2]^T$

Example

- We have

$$A^2 = AA = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^3 = A^2 A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

- And, $A^3 = A^2A = AA^2 = I_2$
- *So,* $A^{-1} = A^2$

Example

$$AX = B \Leftrightarrow A^{-1}AX = A^{-1}B$$

$$\Leftrightarrow X = A^{-1}B = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Theorem 2

Suppose **AX=B** is a system of n equations in n variables and A is an **invertible** matrix. Then the system has the **unique solution**

$$X=A^{-1}B$$

The inverse of an 2x2 matrix

- Consider the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
- The **determinant** of A is **detA=ad-bc**
- **The adjugate matrix of A is defined by** $adjA = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$A^{-1} = \frac{1}{\det A} adjA = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The inverse of 2x2 matrix

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

then $A^{-1} = \frac{1}{\det A} adjA = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$A = \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix}$

$adjA = \begin{bmatrix} -1 & -2 \\ 1 & 3 \end{bmatrix}$

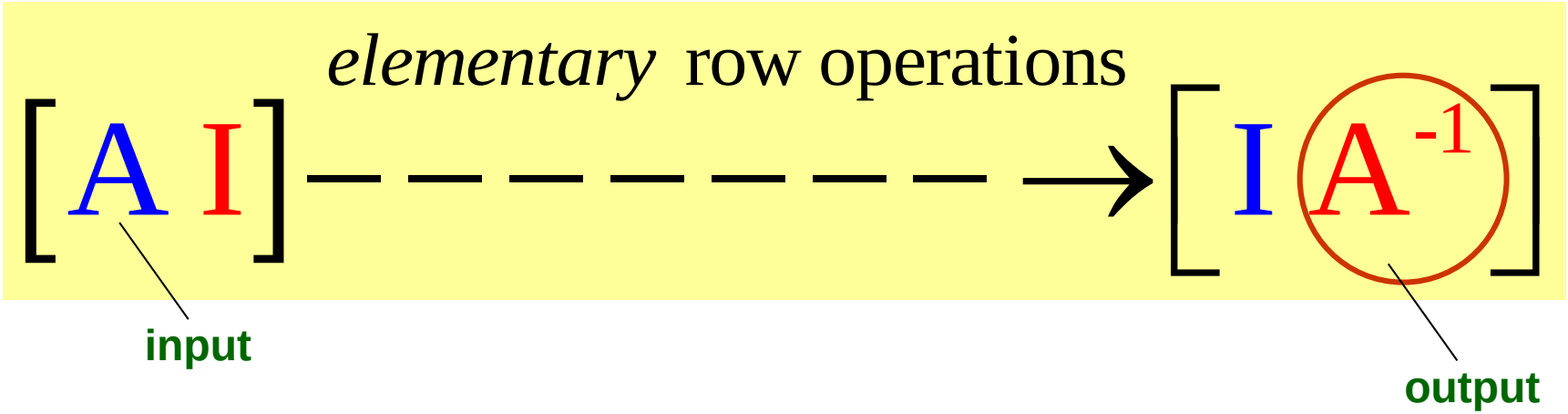
$A^{-1} = \frac{1}{-1} \begin{bmatrix} -1 & -2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix}$

Find the inverse of the following matrices

$$A = \begin{bmatrix} 4 & 3 \\ 6 & 5 \end{bmatrix} \qquad B = \begin{bmatrix} -2 & 1 \\ 7 & 5 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix} \qquad D = \begin{bmatrix} a & 2 \\ 1 & a \end{bmatrix}$$

Matrix Inversion Algorithm



Theorem 3.

Either any square matrix can be reduced to **I** or not.

In the first case, the algorithm produces $\mathbf{A^{-1}}$;

in the second, $\mathbf{A^{-1}}$ does not exist.

Example

- Find the **inverse** of the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$

Example

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 3 & 2 & 0 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow[r_3 \to r_3 + r_1]{r_2 \to r_2 - 3r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 2 & 3 & -3 & 1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{r_2 \leftrightarrow r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 & 0 & 1 \\ 0 & 2 & 3 & -3 & 1 & 0 \end{array}\right]$$

$$\xrightarrow{r_2 \to -r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & -1 \\ 0 & 2 & 3 & -3 & 1 & 0 \end{array}\right] \xrightarrow{r_3 \to r_3 - 2r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array}\right]$$

Example

$$r_2 \to r_2 - r_3$$
$$r_1 \to r_1 + r_3$$
$$\to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & -1 & -3 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 0 & -1 & -3 \\ -1 & 1 & 2 \end{bmatrix}$$

Properties

Theorem 4

- $(A^{-1})^{-1}=A$
- $(AB)^{-1}=B^{-1}A^{-1}$
- $(A^T)^{-1}=(A^{-1})^T$
- $(A_1A_2...A_k)^{-1}=A_k^{-1}...A_2^{-1}A_1^{-1}$
- $(A^k)^{-1}=(A^{-1})^k$
- $(aA)^{-1}=A^{-1}/a$
- $I^{-1}=I$

Corollary. A square matrix A is invertible **iff** A^T is invertible

Example

- Find A if

$$\left(A^T - 2I\right)^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

Example

- Find A if

$$\left(A^T - 2I\right)^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

Theorem 5

The following conditions are equivalent for an **nxn** matrix A

- 1. A is **invertible**
- 2. The homogeneous system AX=0 has **only the trivial solution X=0**
- 3. A can be carried to I_n by elementary row operations
- 4. The system AX=B has **unique solution** for every choice of column B
- 5. There exist an nxn matrix C such that AC=I_n

2.5. Matrix Transformations

(phép biến đổi ma trận)

Definitions

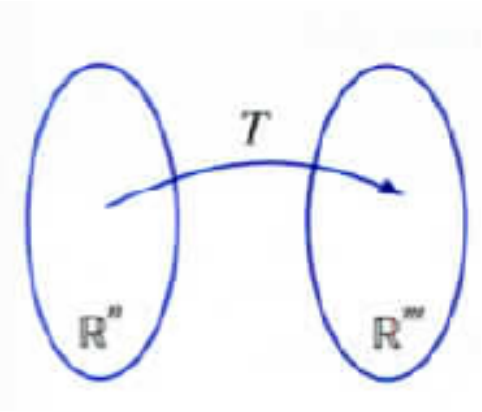
- $(a_1,a_2,\ldots,a_n)$: ordered n-tuple
- R^n: the set of all ordered n-tuples
- Every $(a_1,a_2,\ldots,a_n)$ in R^n is called a vector or n-vector

$$\begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_n \end{bmatrix}$$

Transformations

R^n $\xrightarrow{T}$ R^m

X $\longrightarrow$ $T(X)$



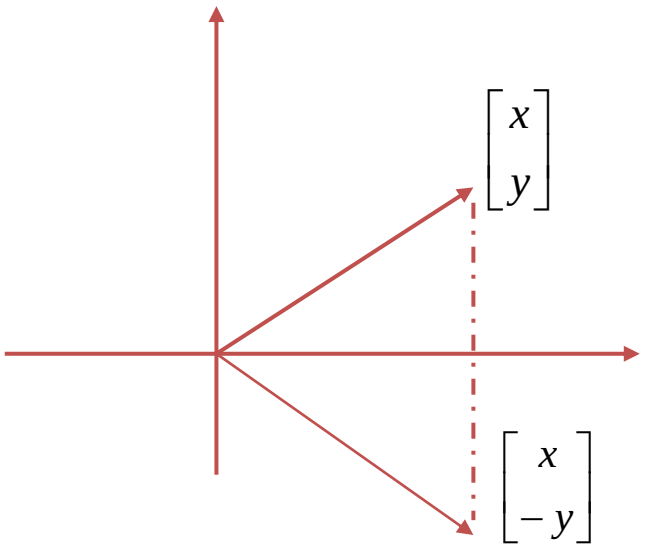
Two transformations T and S are called **equal** iff

$T(X)=S(X)$, for all X in R^n

Transformations

Example 1: Consider the geometricla transformation of R^2 given by reflection in the X axis. This operation carries the vector $\begin{bmatrix} x \\ y \end{bmatrix}$ to its reflection $\begin{bmatrix} x \\ -y \end{bmatrix}$, and so can be describle geometrically as in the diagram. Now observe that

$\begin{bmatrix} x \\ -y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ so reflecting $\begin{bmatrix} x \\ y \end{bmatrix}$ in the X axis can be achieved by multiplying by the matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.



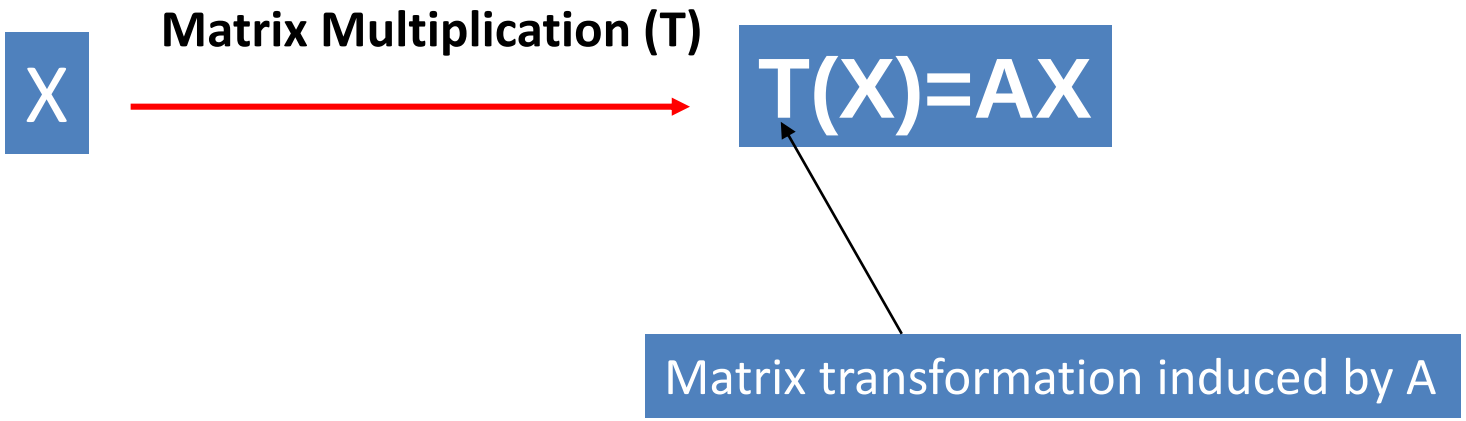
Formula of the Transformation

- $T[x_1 \quad x_2 \quad x_3]^T=[x_1 \quad x_1-x_2 \quad x_2-x_3 \quad x_3-x_1]^T$ defines a transformation $T:R^3 \rightarrow R^4$
- Suppose A is an mxn matrix, then

$T[x_1 \quad x_2 \quad \dots \quad x_n]^T=A[x_1 \quad x_2 \quad \dots \quad x_n]^T$ (**T(X)=AX**) is a transformation **T: $R^n \rightarrow R^m$** called the matrix transformation induced by A

- A is zero matrix: the zero transformation T=0
- A is identity matrix I_n: the identity transformation T= $1_{\mathbb{R}^n}$

Transformation induced by a matrix A



$$0 : \mathbb{R}^n \to \mathbb{R}^m$$

$$X \mapsto 0X = 0$$

T(X)=0X=0

$$1_{\mathbb{R}^n} : \mathbb{R}^n \to \mathbb{R}^m$$

$$X \mapsto I_n X = X$$

T(X)=IX=X

Example

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & 5 \end{bmatrix}$$

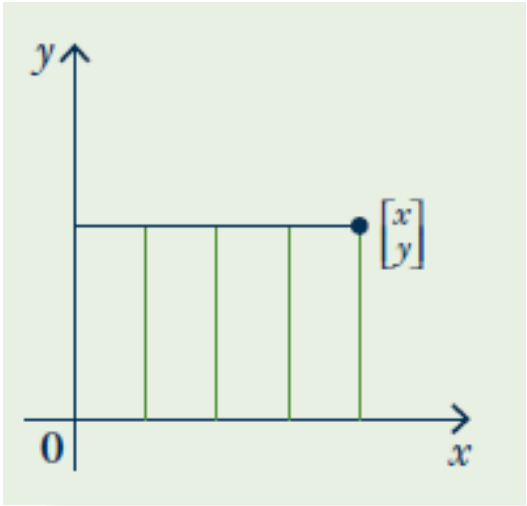
$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \xrightarrow{\text{T(X)=AX}} \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - 2x_2 + 4x_3 \\ 0x_1 + 3x_2 + 5x_3 \end{bmatrix}_{2\times 1}$$

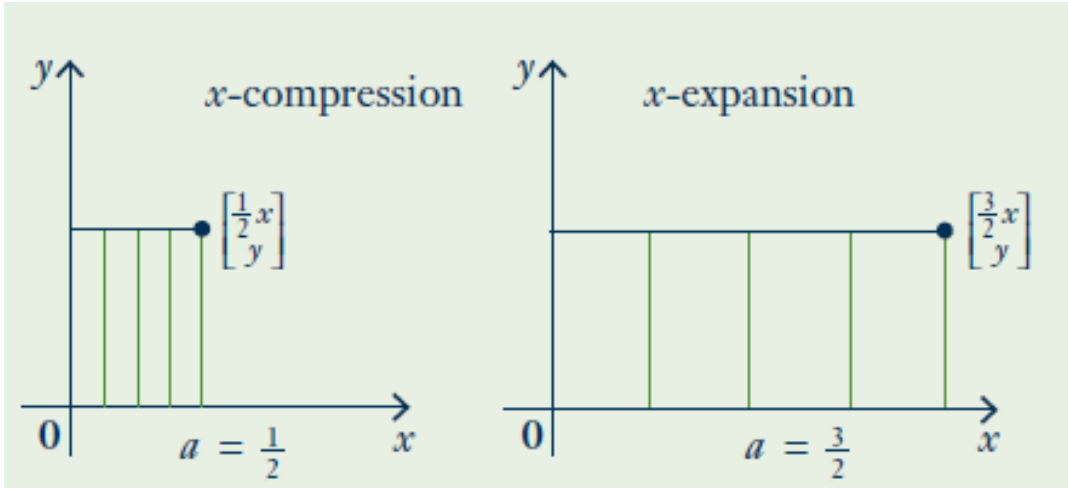
Example

$$A=\begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix} \qquad X=\begin{bmatrix} x \\ y \end{bmatrix}$$

- a>1:**X-expansion** of R^2
- 0<a<1:**X-compression** of R^2

$$\begin{bmatrix} x \\ y \end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} ax \\ y \end{bmatrix}$$





Example

$$A=\begin{bmatrix}1 & 0\\ 0 & b\end{bmatrix} \qquad X=\begin{bmatrix}x\\ y\end{bmatrix}$$

$$\begin{bmatrix}x\\ y\end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix}1 & 0\\ 0 & b\end{bmatrix}\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}x\\ by\end{bmatrix}$$

- b>1:**Y-expansion** of R^2
- 0<b<1:Y-**compression** of R^2

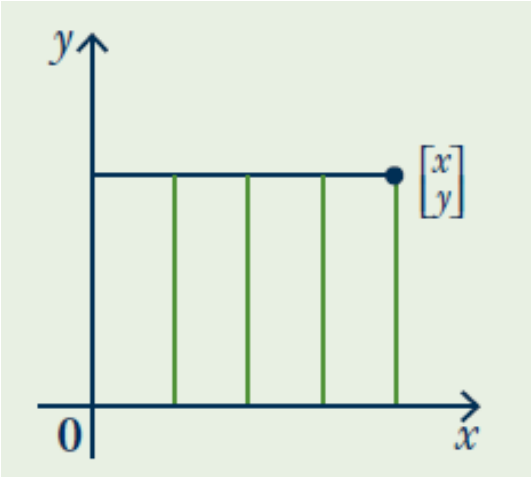
Example

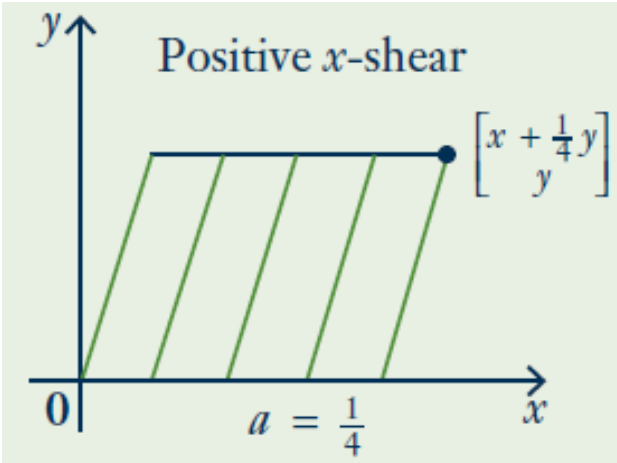
$$A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}$$

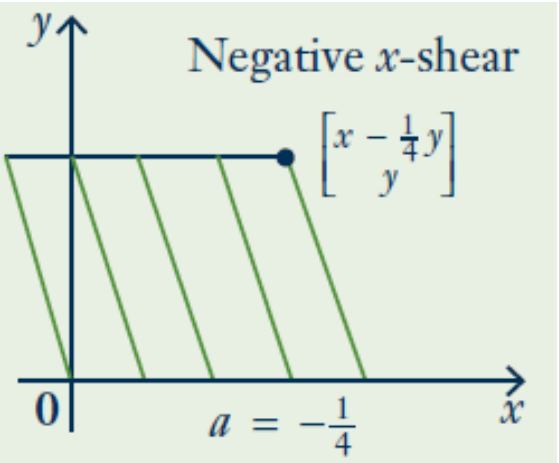
- a>0: Positive **X-shear** of R^2

$$\begin{bmatrix} x \\ y \end{bmatrix} \overset{TX=AX}{\longmapsto} \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+ay \\ y \end{bmatrix}$$

- a<0: Negative **X-shear** of R^2







Example

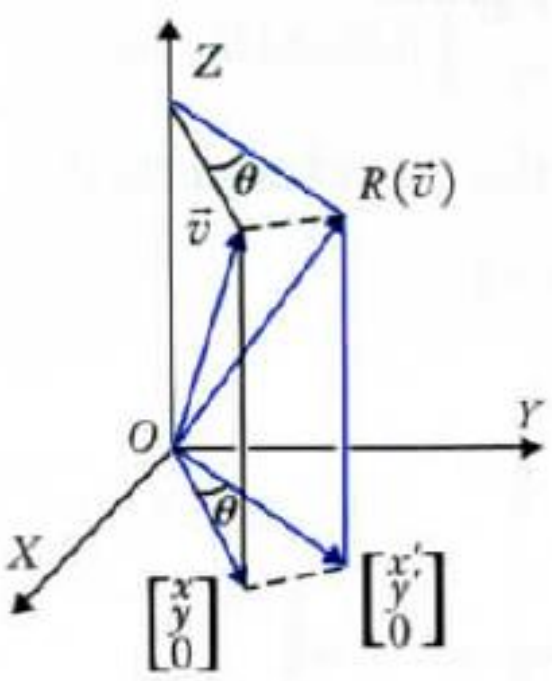
- R_θ: the matrix transformation that **rotates** R^2 counterclockwise (ngược chiều KĐH) about the origin through the angle θ

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \longmapsto \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example

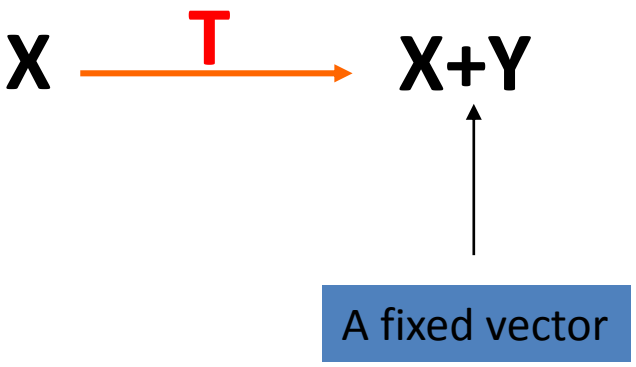
- The rotation of R^3 about the Z axis through an angle θ from the X axis to Y axis

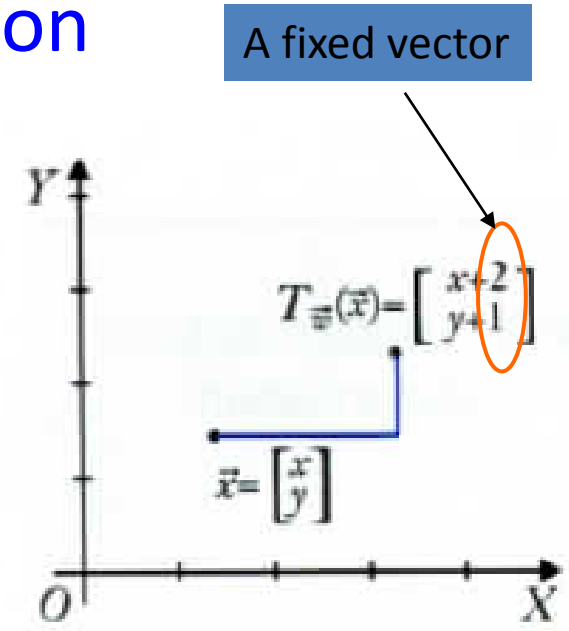


$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation

- There are transformation that are **not** matrix transformations
- For example, the **translation (phép tịnh tiến)** is not a matrix transformation





The matrix does not exist if Y is not zero vector

2.6. Linear Transformation

T: $R^n \rightarrow R^m$ is called a **linear transformation** (phép biến đổi tuyến tính) if it satisfies:

- **1. $T(X+Y)=T(X)+T(Y)$** for all vectors X and Y
- 2. $T(aX)=aT(X)$ for all vector X and all scalar a

If T is a linear transformation then

- $T(0)=0$
- $T(-X)=-T(X)$

Linear Combination
(tổ hợp tuyến tính)

- $Y=a_1X_1+a_2X_2+\ldots+a_nX_n$ is called a **linear combination** of vectors $X_1,X_2,\ldots,X_n$

- **Theorem 1.** If $T: R^n \rightarrow R^m$ is linear transformation, then

$$T(a_1X_1+a_2X_2+\ldots+a_nX_n)$$
$$= a_1 T(X_1)+a_2T(X_2)+\ldots+a_nT(X_n)$$

for all vectors X_i, and all scalar a_i

Linear transformation

Let T:$R^2 \rightarrow R^2$ be linear transformation such that

$$T\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-2\\3\end{bmatrix}, T\begin{bmatrix}3\\0\end{bmatrix}=\begin{bmatrix}4\\5\end{bmatrix}$$

Find $T\left(2\begin{bmatrix}1\\-1\end{bmatrix}+3\begin{bmatrix}3\\0\end{bmatrix}\right)$ and $T\begin{bmatrix}5\\-2\end{bmatrix}$

Linear transformation

Example

$$\begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$T\begin{bmatrix} 5 \\ -2 \end{bmatrix} = T\left(\begin{bmatrix} 3 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$$

$$= T\begin{bmatrix} 3 \\ 0 \end{bmatrix} + 2T\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} + 2\begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

How to find the matrix of an linear transformation?

Theorem 2. Let T: $R^n \rightarrow R^m$ be a transformation

- 1. T is linear iff it is a matrix transformation
- 2. If T is linear, then T is induced by a unique matrix A, given in terms of its columns by $A=[T(E_1) \quad T(E_2) \quad \dots \quad T(E_n)]$ where $E_1, E_2, \dots, E_n$ is the **standard basis** of R^n

Linear transformation

Example

Let $T:R^2 \rightarrow R^2$ be linear such that

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1x - 1y \\ 1x + 3y \end{bmatrix}$$

Find the matrix of T and find $T\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

Example

Standard basis of R^2

$$T\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}1-0\\1+3.0\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix} \quad T\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}0-1\\0+3.1\end{bmatrix}=\begin{bmatrix}-1\\3\end{bmatrix}$$

$$A=\begin{bmatrix}1 & -1\\1 & 3\end{bmatrix} \qquad T\begin{bmatrix}2\\-1\end{bmatrix}=\begin{bmatrix}1 & -1\\1 & 3\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix}=\begin{bmatrix}3\\-1\end{bmatrix}$$

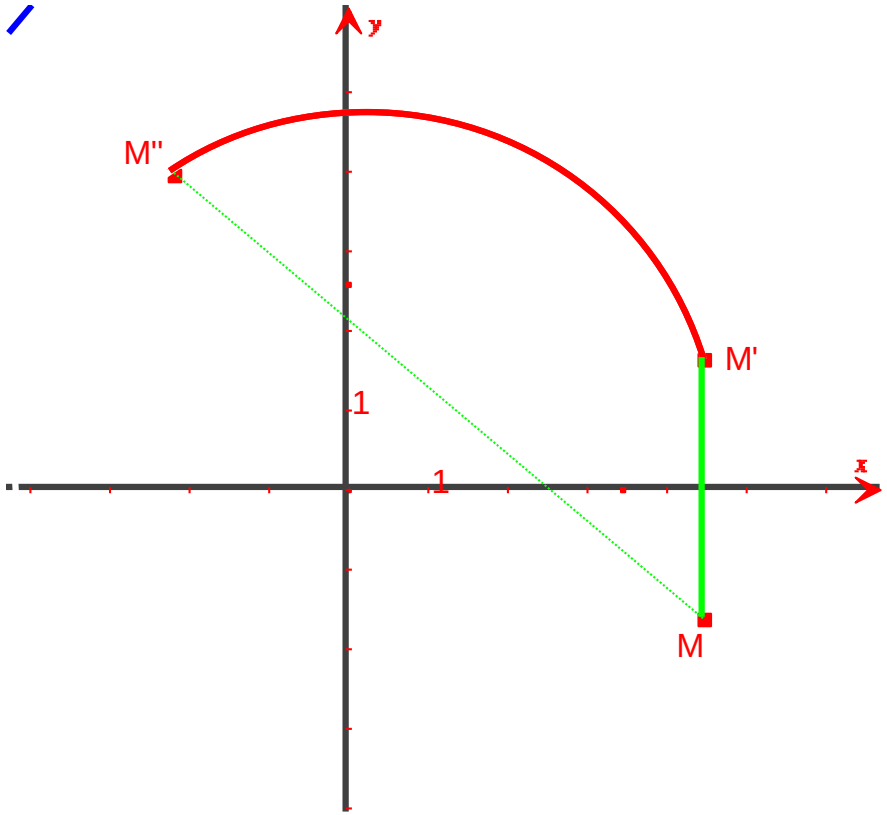
Composition

- $T:R^n \rightarrow R^m$ and $S:R^m \rightarrow R^k$
- The **composite** of S and T is a transformation defined by $(S \circ T)(X)=S[T(X)]$ for all vector X in R^n
- **Theorem 3.** If T and S are linear then $S \circ T$ is also linear. And if S has matrix A and T has matrix B, then $S \circ T$ has matrix AB

Transformation	**S**	**T**	**$S \circ T$**
Matrix	**A**	**B**	**AB**

Example

- The **reflection** in the X axis followed by **rotation** through **pi/2** is **reflection** in the line y=x (ex. 9 p.78)

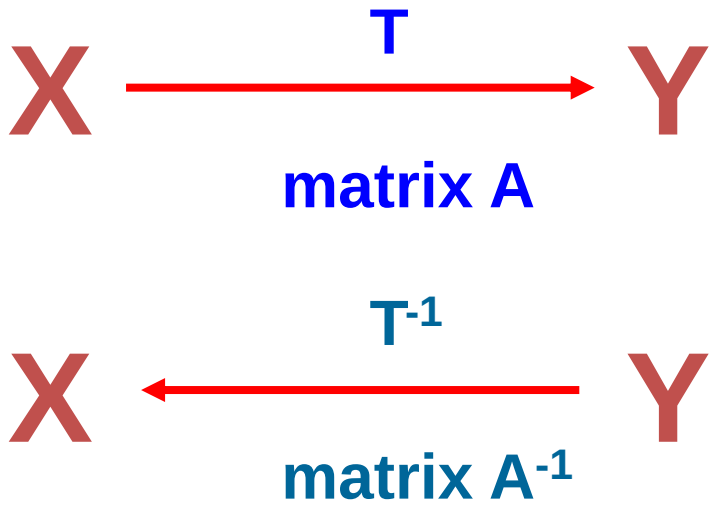


Inverse

- T:$R^n \rightarrow R^n$ is a transformation that has matrix A
- Suppose A is **invertible** then there exist A^{-1}
- Let T':$R^n \rightarrow R^n$ is a transformation that has matrix A^{-1}
- $T'[T(X)]=A^{-1}[AX]=IX=X$
- $T[T'(X)]=A[A^{-1}X]=IX=X$
- T' is called an **inverse** of T if $T \circ T' = 1_{\mathbb{R}^n} = T' \circ T$

Inverse

Theorem 5. Let $T:R^n \rightarrow R^n$ denote a linear transformation with matrix A. Then A is invertible if and only if T has an inverse. In this case, T is has **exactly one** inverse (which we denote as T^{-1}) and T^{-1} has matrix A^{-1}



Linear transformation

Example

Let $T:R^2 \rightarrow R^2$ be a linear such that

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x-y \\ x+3y \end{bmatrix}$$

Find the matrix of T and find $T^{-1}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

Example

$$A=\begin{bmatrix}1 & -1\\ 1 & 3\end{bmatrix}\Rightarrow A^{-1}=\frac{1}{4}\begin{bmatrix}3 & 1\\ -1 & 1\end{bmatrix}$$

$$T^{-1}\begin{bmatrix}2\\ -1\end{bmatrix}=A^{-1}\begin{bmatrix}2\\ -1\end{bmatrix}=\frac{1}{4}\begin{bmatrix}3 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}2\\ -1\end{bmatrix}=\frac{1}{4}\begin{bmatrix}5\\ -3\end{bmatrix}$$

P40: 3,4,15

P67: 2-5, 11-13, 25, 26

P80: 1-6, 16,17

P101: 1-3,8,12